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On the glass transition domain in some M-SiAlON (M = Y or Ln) oxynitride glasses

J.-L. Besson ^{a,*}, G. Massouras ^a, A. Bondanini ^a, M. Huger ^a, S. Hampshire ^b,
Y. Menke ^b, H. Lemerrier ^b

^a ENSCI, 47 Av. A. Thomas, 87065 Limoges cedex, France

^b Materials and Surface Science Institute, University of Limerick, Ireland

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Abstract

The behaviour in the glass transition domain of some oxynitride glasses has been studied by thermoanalytical methods (dilatometry and differential thermal analysis) and mechanical techniques (creep and ultrasonic measurements of Young's modulus). The thermoanalytical data are in good agreement with the glass transition domain defined from viscosity data. The sharp decrease of Young's modulus, that starts at the temperature of the strain point, is compared to results obtained from mechanical spectroscopy by other authors. The difference in the apparent activation energies for viscous flow above and below the temperature of the strain point is used to separate the contribution of the thermal and structural components. The high temperature apparent activation energy of viscosity is in fair agreement with that of the α -relaxation peak described by the formalism of hierarchically correlated molecular mobility. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

The glass transition (transition between the metastable equilibrium state of an undercooled liquid and the non-equilibrium state of a glass or a glassy solid) occurs when the system loses the ability of reaching the configuration corresponding to the minimum free enthalpy. This happens when the characteristic time for configurational change becomes large compared to the experimental time for cooling. Through the glass transition domain, the second derivatives of the free

enthalpy (heat capacity and expansion coefficient) exhibit a more or less sudden change from liquid-like to crystal-like values. Upon heating, the change giving rise to the transition is the result of the appearance of a structural contribution in the measured quantity. Due to kinetic factors and relaxation processes occurring in the region of the transition, its actual temperature depends on the history of the material (initial cooling rate, heating rate, structural ageing) and the experimental conditions used during the measurements.

The fact that the system becomes unable to reach the state of minimum free enthalpy comes from a drastic change in the molecular mobility, and it is common to think of the glassy state in terms of viscous flow process, a glass being viewed as a liquid, the viscosity of which has become

* Corresponding author. Tel.: +33-5 55 45 22 22; fax: +33-5 55 79 09 98.

E-mail address: jl.besson@ensci.fr (J.-L. Besson).

greater than a specified value. The glass transition domain is then defined by the temperature range through which the viscosity changes from $10^{12.6}$ to 10^{12} Pa s.

Relaxation processes due to configurational rearrangements of atomic or molecular constituents of the glass can be induced by an external applied field (electric or mechanical). Of particular interest is the so-called principal relaxation (or α -relaxation) which is related to the onset of long-range molecular movements. This relaxation appears when the material is submitted to a cyclic stress. It is thermally activated and is characterised by a dissipation peak in the imaginary part of the elastic modulus, associated with a strong decrease in the real part of the modulus. The peak maximum appears at a temperature near T_g when the frequency of the cyclic stress is around 1 kHz and shifts to higher temperature for higher frequency.

The elastic moduli measured by high frequency methods decrease weakly from room temperature up to near T_g and the decrease is more pronounced for higher temperatures.

The purpose of the present paper is to compare the transition temperatures obtained by thermo-analytical methods (dilatometry, differential thermal analysis), creep measurements of the viscosity and ultrasonic measurements of Young's modulus in some oxynitride glasses in MSiAlON systems with $M = Y, La, Eu$ and Er . The change through the transition domain in the apparent activation energy for viscosity is also discussed.

2. Materials

Bulk glasses were prepared from mixtures of adequate amounts of high purity oxides and silicon nitride powders to get the desired chemical compositions. Y_2O_3 and the rare earth oxides with 99.9% purity were from Rare Earth Products, Al_2O_3 was analar grade (BDH), SiO_2 was from Cerac (99.5%) and Si_3N_4 from Starck (LC12SX). All oxides were calcined to remove any volatiles and/or chemically absorbed water. The powders were wet ball milled in isopropanol for 24 h, followed by evaporation of the alcohol before pressing into pellets. Large batches (50–60 g) were melted at 1700°C in a boron nitride lined graphite crucible under 0.1 MPa nitrogen atmosphere in a vertical tube furnace after which the melt was poured into a preheated graphite mould at 850°C and annealed for 1 h prior to slow furnace cooling to stabilise the glasses. The nominal compositions in equivalent percent (e/o) [1] and atomic percent are given in Table 1.

3. Experimental techniques

3.1. Dilatometry

Thermal expansion coefficients were measured in a flowing nitrogen atmosphere from room temperature up to temperatures above the dilatometric softening point with a heating rate of 5 K/min. The samples were parallelepipedic bars ap-

Table 1
Nominal compositions in equivalent and atomic concentrations of the various compositions

	Cation (e/o)			Anion (e/o)		Cation (at%)			Anion (at%)	
	Y	Si	Al	O	N	Y	Si	Al	O	N
YSi17G16	28	56	16	83	17	12.3	18.5	7.0	54.7	7.5
Y-B phase	35	45	20	83	17	15.1	14.7	8.7	54.7	7.4
Y-Q phase	18	47	35	80	20	7.9	15.4	15.3	52.6	8.8
La						La				
La-W phase	21.4	57.2	21.4	87	13	9.35	18.7	9.35	57.0	5.6
Er						Er				
Er-B phase	35	45	20	83	17	15.1	14.7	8.7	54.7	7.4
Eu						Eu				
Eu ₃ Si ₆ Al ₆ O ₂₁ N ₂	12.5	50	37.5	87.5	12.5	7.9	15.8	15.8	55.3	5.2

proximately $3 \times 3 \times 10 \text{ mm}^3$. The mean values of the thermal expansion coefficients were calculated between 20°C and 800°C where changes are linear.

3.2. Differential thermal analysis

Differential thermal analysis (DTA) was carried out in order to detect the glass transition temperature (an endothermic change) and crystallisation temperature (an exothermic peak). The instrument used was a Stanton Redcroft 780 series simultaneous thermo-gravimetric differential analyser. Small samples (60 mg) were analysed in boron nitride lined platinum crucibles in a flowing nitrogen atmosphere. Alumina was used as a reference material. The heating rate was the same as for the dilatometric measurements: 5 K/min.

3.3. Viscosity

Viscosity, η , was deduced from compressive creep experiments performed in air with temperature jumps between 800°C and 1000°C, using the well-known analogy between pure elastic (Hookean) and pure viscous (Newtonian) behaviours

$$\eta = \sigma / [2(1 + \nu)\dot{\epsilon}],$$

where $\dot{\epsilon}$ and σ are the steady-strain rate and flowing stress, respectively, and ν is Poisson's ratio. The apparatus has been calibrated using a commercial silica glass (Corning 'Vision'), the viscosity of which is well documented. The sample size was $3 \times 3 \times 7 \text{ mm}^3$. To minimise the effect of structural relaxation during creep, the temperature jumps were done as soon as a stationary regime was seen to be established.

3.4. Elastic moduli

At room temperature, the elastic moduli (E , G) were calculated from the measurements of the longitudinal, V_l , and transverse, V_t , ultrasonic wave velocities by means of 10 MHz piezoelectric transducers using the 'Pulse Echo Overlap' technique. E , G and ν are derived from the classical relationships for propagation in an infinite medium:

$$E = \rho[3V_l^2 - 4V_t^2] / [(V_l^2/V_t^2) - 1],$$

$$G = \rho V_t^2,$$

$$\nu = (E/2G) - 1,$$

where ρ is the density of the material.

For measurements at high temperature, the magnetostrictive transducer is kept at room temperature and sealed to the sample via a refractory guide. The experimental device has been described elsewhere [2,3]. The technique uses low frequency pulses (300 kHz) and the data analysis is based on the 'long bar mode' approximation. For longitudinal wave,

$$E = \rho V_l^2.$$

The samples were parallelepipedic bars $2 \times 2 \times 35 \text{ mm}^3$.

4. Results and discussion

An illustration of the experimental curves is given in Figs. 1–3 for the YSi17G16 glass.

In the case of the dilatometric measurements, T_g was taken as the extrapolated onset temperature of the glass transition. The evaluation of T_g from DTA experiment can be made by different procedures [4]. As the endothermal shift is generally small in these oxynitride glasses [5], the tempera-

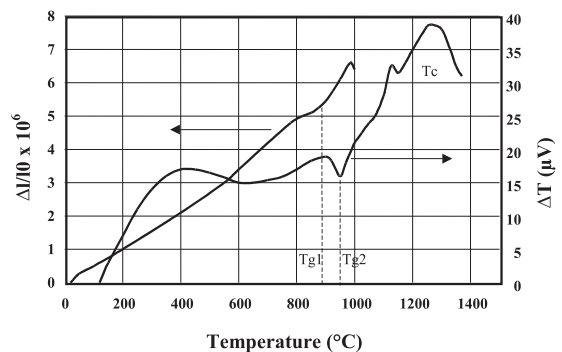


Fig. 1. Dilatometric curve and DTA trace for YSi17G16 glass. Heating rate 5°C/min. T_{g1} corresponds to the onset of the glass transition domain obtained by dilatometric measurements; T_{g2} to a temperature closed to the end of the glass transition domain and T_c is the temperature of the exothermal crystallisation peak.

ture of the minimum on the DTA trace was selected as T_g . It corresponds to a temperature close to the end of the transition. The entire set of results

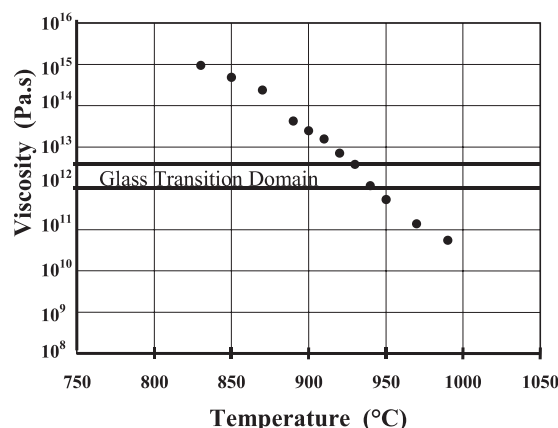


Fig. 2. Viscosity versus temperature curve for YSi17G16 glass.

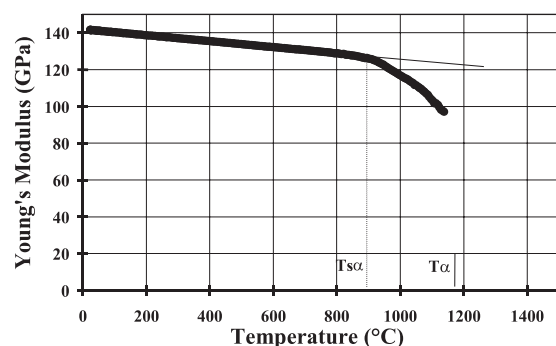


Fig. 3. Change in Young's modulus with temperature for YSi17G16 glass. T_α is the extrapolated temperature of the mechanical loss peak at 300 kHz, $T_{s\alpha}$ the temperature corresponding to the onset of the associated decrease in the real part of the elastic modulus.

is collected in Table 2. As can be seen from this table, T_g from dilatometry and T_g from DTA are reasonably consistent with the lower and upper limits of the glass-transition as defined by the characteristic points ($\eta = 10^{12.6}$ and 10^{12} Pa s, respectively) deduced from the viscosity curves.

On the other hand, the transition temperature determined from the change in Young's modulus is systematically lower and corresponds to a viscosity of about $10^{13.5}$ Pa s, that is close to the strain point.

Donzel et al. [6], in Lausanne (Switzerland), have carried out a detailed study on the same YSi17G16 glass, provided by the University of Limerick, by mechanical spectroscopy using a differential forced inverted pendulum working at frequencies between 10^{-3} and 10 Hz. Their measurements allowed them to calculate the real and imaginary parts of the dynamical modulus: $G^* = G' + iG''$. The spectra showed above T_g a mechanical peak loss in $G''(T)/G_u$ associated with a strong decrease in $G'(T)/G_u$ (G_u is the unrelaxed modulus) which they attributed to the α -relaxation. The usual representation of the tangent of the mechanical loss angle, $\tan \Phi = G''/G'$, or internal friction, is not appropriate in the case of inorganic glasses because at high temperature the modulus drops to zero and $\tan \Phi$ shows an exponential increase instead of a peak. Such an exponential increase above T_g has also been reported by Duffrène et al. [7] in their study of the high frequency viscoelastic behaviour of a soda-lime-silica glass using the Brillouin scattering technique (frequency ~ 20 GHz). The start of the exponential increase was associated with a decrease in the elastic moduli. Using the same technique, Xu and Manghanani [8] also have observed a turning point

Table 2

Elastic constants at room temperature, thermal expansion coefficient and transition temperatures derived from the different techniques^a

	E (GPa)	G (GPa)	ν	α (10^{-6} K^{-1})	T_g dilato. (°C)	T_g ATD (°C)	T_g viscosity (°C)	T_{us} (°C)
YSi17G16	142	55	0.30	6.1 ± 0.2	935	943	930–945	925
Y–B phase	148	58	0.30	7.1 ± 0.2	932	950	935–950	890
Y–Q phase	148	58	0.29	3.4 ± 0.2	928	948	925–942	904
La–W phase	129	50	0.29	3.95 ± 0.2	883	907	880–905	866
Er–B phase	154	60	0.29	6.7 ± 0.2	926	955	919–940	858
$\text{Eu}_3\text{Si}_6\text{Al}_6\text{O}_{21}\text{N}_2$	93	37	0.27	6.35 ± 0.2	909	929	890–915	861

^a The accuracy on the elastic moduli is 1% and the temperature are determined at $\pm 2^\circ\text{C}$.

at the strain point in the velocities of the shear and longitudinal modes in a sodium silicate glass. They reported that their experimental data were in good agreement with results obtained by Wang and Manghanani from ultrasonic measurements. They concluded that a structural change should occur at the strain point. Then, we suggest that the transition temperature observed in the change in slope of Young's modulus in our ultrasonic measurements is to be identified with the onset of the modulus defect associated with the α -relaxation.

The maximum of the mechanical loss, T_α , was found by Donzel et al. at 1025°C for a measurement frequency of 1 Hz. The apparent activation

energy was reported to be 1116 ± 20 kJ/mol. Extrapolating their measurements for a frequency of 300 kHz would give a temperature of 1187°C, but at this temperature the glass crystallises [9] (see Fig. 1). It is the reason why we focused on the onset of the change in elastic modulus and not on the maximum of the mechanical loss.

Plots of the logarithm of the viscosity versus the reciprocal absolute temperature present clearly, two different slopes for all the studied glasses (Fig. 4). The intersection of the two straight lines occurs always at the transition temperature observed on Young's modulus curves, i.e., for a viscosity corresponding to the strain point. The

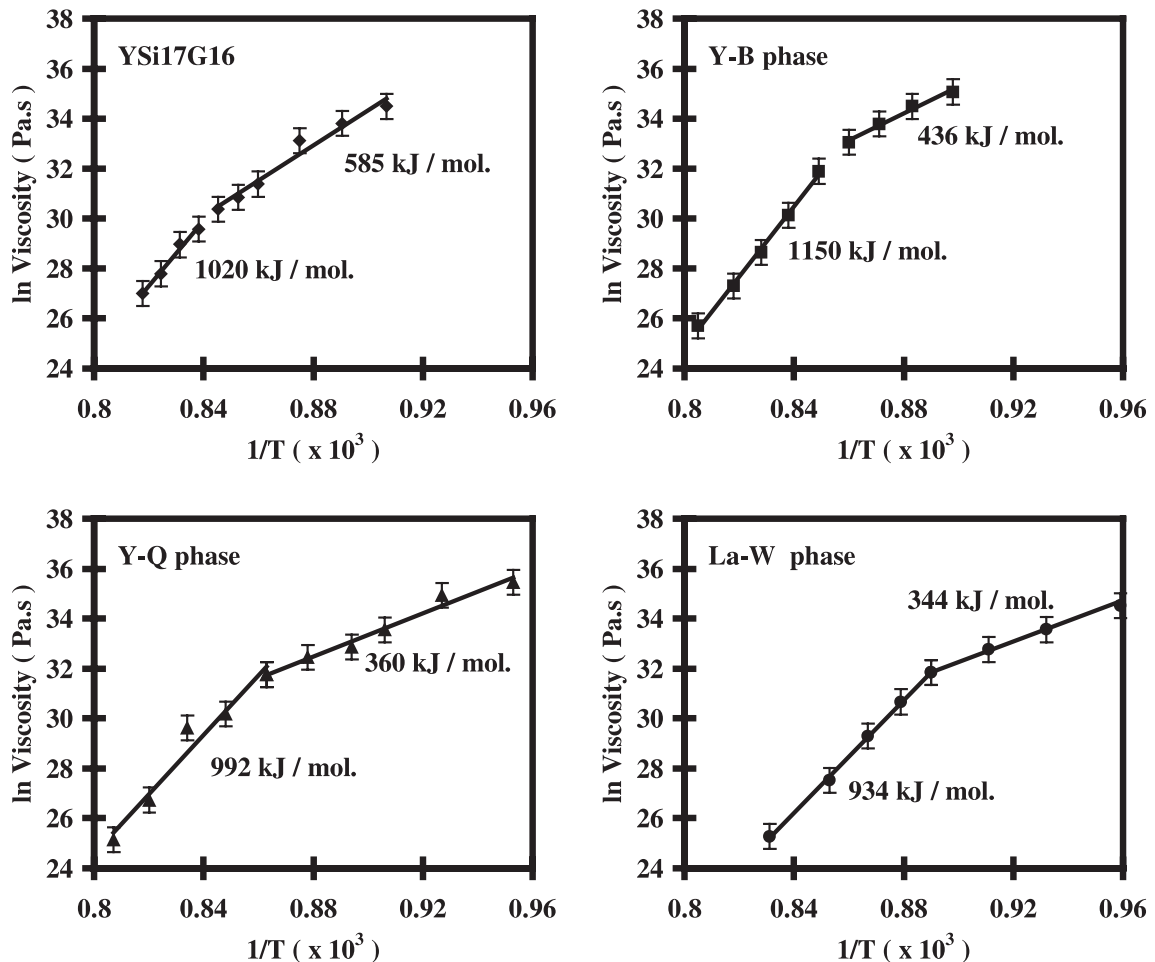


Fig. 4. Apparent activation energies.

Table 3
Apparent activation energies^a

H (kJ/mol)	H_a $T > T_{us}$	$H_a = H_{therm.}$ $T < T_{us}$	$H_{struct.}$ $T > T_{us}$
YSi17G16	1020	585	435
Y–B phase	1150	436	714
Y–Q phase	992	360	632
La–W phase	934	344	590
Er–B phase	934	264	670
Eu ₃ Si ₆ Al ₆ O ₂₁ N ₂	905	284	621

^a The accuracy on the measurement of H_a is ± 50 kJ/mol.

values of the apparent activation energies for $T < T_{us}$ and $T > T_{us}$ are reported in Table 3.

At high temperature, any change in temperature results in a change in the translational molecular mobility coming from two independent effects: one corresponding to the probability of occurrence of thermally activated events, the other to changes in the configurational organisation. As the temperature decreases, the specific volume decreases, the entropy decreases (i.e., the order increases), the intermolecular bonds become stronger. Whatever the theoretical approach (free volume [10,11], entropy fluctuation [12] or quasi-point defects [13,14], the same conclusion is reached: it is more and more difficult to activate the translational degrees of freedom. At low temperature, changes in temperature only results in changes in the thermal activation, the configurational arrangement of the structural units being frozen in. So, the apparent activation energy may be written as the sum of thermal and structural components

$$H_a = H_{therm.} + H_{struct.},$$

$H_{struct.}$ being zero at $T < T_{us}$. Then, the difference between the apparent activation energies on both sides of T_{us} gives the structural component.

In the case of the YSi17G16 glass the apparent activation energy at high temperature (1020 kJ/mol) is similar to that determined by Donzel et al. for the α -relaxation. They interpret their results in the frame of the molecular theory proposed by Perez et al. [13,14]. These authors analyse the rheological behaviour of amorphous materials in terms of nucleation and expansion of sheared microdomains under an applied stress. The sheared microdomains are initiated at sites of

density fluctuations, called quasi-point defects, in the glass network. The expansion of the sheared microdomains results from a ‘hierarchically correlated motion’, originally introduced by Palmer et al. [15], i.e., the faster movements happen first and enable the occurrence of slower ones in a hierarchic series. The importance of the hierarchical correlation upon mobility is expressed by a parameter b , $0 < b < 1$, proportional to the concentration of quasi-point defects: the lower the value of b , the stronger the correlation effects and the more difficult the deformation. b is a constant in the non-equilibrium glassy state and is a function of temperature in the supercooled liquid state.

For the YSi17G16 glass, b was found equal to 0.48, a rather high value consistent with the high thermomechanical properties of yttrium oxynitride glasses. Moreover, b was temperature independent in the 830–1030°C temperature range, so that the glass showed an Arrhenian behaviour above T_g . Under these conditions, the activation enthalpy, $H(T)$, of the relaxation time is independent of the temperature and is related to the free activation energy, G , of the elementary molecular motion by: $H = G/b$, that leads to a value of $G \approx 530$ kJ/mol. This value is in reasonable agreement with the value of 470 kJ/mol obtained in an early work [16] from creep experiments, following the thermodynamic and kinetic analysis of non-elastic deformation in polymeric glasses developed by Esgaig and co-workers [17,18] by extending the thermodynamics and kinetics of activated plasticity established for crystals [19,20].

5. Conclusion

The behaviour of different oxynitride glasses has been studied through the glass transition domain by various techniques. The values for T_g derived from dilatometry (onset of the transition) and DTA (end of the transition) were in good agreement with the temperature range defined by viscosity data.

It has been shown that the sharp decrease of Young’s modulus, measured using an ultrasonic technique at 300 kHz, starts at a temperature

corresponding to the strain point and is due to the onset of the α -relaxation.

Values for the thermal and structural components of the apparent activation energy for viscous flow have been obtained from the change in slope in the $\ln$ viscosity versus reciprocal temperature curves, above and below the temperature of the strain point.

The thermoanalytical data derived from creep tests for the YSi17G16 glass have been shown to be in fair agreement with the values obtained on the same glass by other researchers using mechanical spectroscopy.

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